



Testing and estimating structural breaks in time series and panel data in Stata

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Abstract. Identifying structural change is a crucial step when analyzing time series and panel data. The longer the time span, the higher the likelihood that the model parameters have changed because of major disruptive events such as the 2007–2008 financial crisis and the 2020 COVID-19 outbreak. Detecting the existence of breaks and dating them is therefore necessary for not only estimation but also understanding drivers of change and their effect on relationships. In this article, we introduce a new community-contributed command called `xtbreak`, which provides researchers with a complete toolbox for analyzing multiple structural breaks in time series and panel data. `xtbreak` can detect the existence of breaks, determine their number and location, and provide break-date confidence intervals. We use `xtbreak` in examples to explore changes in the relationship between COVID-19 cases and deaths in the US using both aggregate and state-level data and in the relationship between approval ratings and consumer confidence using a panel of eight countries.

Keywords: `st0781`, `xtbreak`, structural breaks, change points, time-series data, panel data, interactive fixed effects, cross-sectional dependence

1 Introduction

In economics and elsewhere, linear relationships between dependent and explanatory variables are at the core of interest. To investigate such relationships, researchers collect observations over time for one or more cross-sectional units such as firms, individuals, or countries and subsequently use them in estimating the coefficients of regression models. A key assumption here is that the coefficients do not change over time. This assumption is unlikely to hold, especially for longer periods of time, because of major disruptive events such as financial crises. Parameter instability can have a detrimental impact on

estimation and inference and can lead to costly errors in decision making. The times in which the parameters change are called “change points” in the statistics literature and “structural breaks” in economics. Because both terms are synonymous to each other, we will use the latter term or just “breaks”.

In this article, we propose a new community-contributed command called **xtbreak**.¹ The package implements the methods developed by Bai and Perron (1998) for pure time series and Ditzen, Karavias, and Westerlund (2025) for panel data.²

xtbreak provides researchers with a complete toolbox for analyzing multiple structural breaks in time series and panel data. It can detect and date an unknown number of breaks at unknown break dates. The toolbox is based on asymptotically valid tests for the presence of breaks, a consistent break-date estimator, and a break date confidence interval with correct asymptotic coverage. In fact, **xtbreak** includes three tests: 1) a test of no structural breaks against the alternative of a specific number of breaks, 2) a test of the null hypothesis of no structural breaks against the alternative of an unknown number of structural breaks, and 3) a test of the null of s breaks against the alternative of $s + 1$ breaks. The package also includes an algorithm that uses the last test consecutively to estimate the true number of breaks. The tested break dates can be unknown or user defined, as when researchers have additional information and wish to examine whether there was a break in a specific point in time. Once the presence of breaks has been tested and confirmed, **xtbreak** estimates the locations of the breaks and provides the associated confidence intervals.

Having many breaks does not translate into heavy computational burden: **xtbreak** implements an efficient dynamic programming method described in Bai and Perron (2003) that ensures that there are $O(T^2)$ computations even with more than two breaks, where T is the number of time-series observations.

xtbreak can deal with models of “pure” or “partial” structural breaks. In a pure structural-breaks model, the coefficients of all explanatory variables change, while in a partial structural-breaks model, only a subset of the coefficients changes.

xtbreak is applicable under very general error conditions. For time-series data, the only requirement is that there are no unit roots in the errors. For panel data, units can be independent or cross-sectionally dependent, where cross-sectional dependence takes an “interactive fixed effects” or “common factor” structure. Regressors can load on the same set of factors as the errors, which means that regressors may be endogenous—although no instrumental variables are necessary. The errors can also be serially correlated and heteroskedastic but not nonstationary.

The works of Bai and Perron (1998) and Ditzen, Karavias, and Westerlund (2025) draw on the economics literature. However, structural breaks also happen in other fields of research, including engineering, epidemiology, climatology, and medicine. **xtbreak** is

1. Updates will be continuously provided on our GitHub page: <https://janditzen.github.io/xtbreak/>.

2. The Ditzen, Karavias, and Westerlund (2025) study develops the methods and asymptotic theory for the analysis of panel data with multiple structural breaks and interactive fixed effects. For one break, the results coincide with the earlier work by Karavias, Narayan, and Westerlund (2023).

therefore widely applicable. To showcase this width, we consider two examples drawn from the areas of epidemiology and political economy. First, we consider the epidemiological relationship between COVID-19 cases and deaths. Using both aggregate country and disaggregated state-level US data, we find evidence of multiple breaks. In particular, we find that an increase in the number of COVID-19 cases led to more deaths in the beginning of the pandemic than in later waves. Second, we examine if there are breaks in the relationship between consumer confidence and the approval ratings of country leaders. Using a panel of eight countries observed over a long period of time, we find that there is great cross-country heterogeneity in terms of the number and locations of breaks.

The remainder of the article is organized as follows: Section 2 presents the model that we will be considering. We focus on the panel case, which in most regards includes the pure time series setup as a special case. Important differences are brought up and discussed. Sections 3, 4, and 5 present the hypothesis tests, the break-date estimation procedure, and the `xtbreak` command, respectively. Sections 6 and 7 contain the empirical analyses of the COVID-19 and leader approval ratings data, respectively. Section 8 concludes the article.

2 Model discussion

We consider the following model with N units, T periods, and s structural breaks:

$$y_{i,t} = \mathbf{x}'_{i,t}\boldsymbol{\beta} + \mathbf{w}'_{i,t}\boldsymbol{\delta}_j + e_{i,t} \quad (1)$$

$t = T_{j-1}, \dots, T_j$ and $j = 1, \dots, s+1$ with $T_0 = 0$ and $T_{s+1} = T$. Hence, there are s breaks, or $s+1$ regimes with regime j covering the observations $T_{j-1}, \dots, T_j$. To emphasize the break structure, we can write (1) regimewise:

$$\begin{aligned} y_{i,t} &= \mathbf{x}'_{i,t}\boldsymbol{\beta} + \mathbf{w}'_{i,t}\boldsymbol{\delta}_1 + e_{i,t} \text{ for } t = T_0, \dots, T_1 \\ y_{i,t} &= \mathbf{x}'_{i,t}\boldsymbol{\beta} + \mathbf{w}'_{i,t}\boldsymbol{\delta}_2 + e_{i,t} \text{ for } t = T_1, \dots, T_2 \\ &\vdots \\ y_{i,t} &= \mathbf{x}'_{i,t}\boldsymbol{\beta} + \mathbf{w}'_{i,t}\boldsymbol{\delta}_{s+1} + e_{i,t} \text{ for } t = T_s, \dots, T_{s+1} \end{aligned}$$

For $N = 1$, this is a time-series model, while for $N > 1$, it is a panel-data model. The dependent variable $y_{i,t}$ and the regression error $e_{i,t}$ are scalars, while $\mathbf{x}_{i,t}$ and $\mathbf{w}_{i,t}$ are $p \times 1$ and $q \times 1$ vectors, respectively, of regressors. The coefficients of the regressors in $\mathbf{x}_{i,t}$ are unaffected by the breaks, while those of $\mathbf{w}_{i,t}$ are affected by the breaks. It is possible that all independent variables break, in which case $\mathbf{x}'_{i,t}\boldsymbol{\beta}$ is defined to be zero. In the panel case, the break dates are common for all units. This common assumption is reasonable when the frequency of the data is not high. Let $\mathcal{T}_s = \{T_1, \dots, T_s\}$ be a collection of s break dates such that $T_j = \lfloor \lambda_j T \rfloor$, where $\lambda_0 = 0 < \lambda_1 < \dots < \lambda_s < \lambda_{s+1} = 1$. By specifying the breaks this way, we ensure that they are distinct from one another and bounded away from the beginning and end of the sample. This is important because we need to be able to fit the model within each regime.

If the data have a panel structure ($N > 1$), we allow for unobserved heterogeneity in the form of interactive fixed effects:

$$e_{i,t} = \mathbf{f}_t' \boldsymbol{\gamma}_i + \varepsilon_{i,t} \quad (2)$$

$\mathbf{f}_t$ is an $m \times 1$ vector of factors, and $\boldsymbol{\gamma}_i$ is a conformable vector of factor loadings.³ The fact that $\mathbf{f}_t$ is common to all cross-sectional units i means that the regression errors can be strongly cross-sectionally correlated. This specification is very general and nests the usual one-way and two-way fixed-effects models as special cases. Both $\mathbf{f}_t$ and $\varepsilon_{i,t}$ may be weakly serially correlated, but they cannot be nonstationary. They can also not be correlated with each other, and $\varepsilon_{i,t}$ cannot be cross-sectionally correlated.⁴ This last condition ensures that any cross-sectional dependence in $e_{i,t}$ originates with $\mathbf{f}_t$.

Typically, there is a lot of cross-sectional comovement not only in the regression errors but also in the regressors. To account for this, we assume that $\mathbf{x}_{i,t}$ and $\mathbf{w}_{i,t}$ are generated as

$$\mathbf{x}_{it} = \boldsymbol{\Gamma}_{x,i}' \mathbf{f}_t + \mathbf{u}_{x,i,t} \quad (3)$$

$$\mathbf{w}_{it} = \boldsymbol{\Gamma}_{w,i}' \mathbf{f}_t + \mathbf{u}_{w,i,t} \quad (4)$$

where $\boldsymbol{\Gamma}_{x,i}$ and $\boldsymbol{\Gamma}_{w,i}$ are $p \times m$ and $q \times m$ matrices, respectively, of factor loadings, while $\mathbf{u}_{x,i,t}$ and $\mathbf{u}_{w,i,t}$ are $p \times 1$ and $q \times 1$ vectors, respectively, of idiosyncratic errors that are independent of all the other random elements of the model. The model described by (1)–(4) above is the same as the one considered by Ditzen, Karavias, and Westerlund (2025).⁵

The fact that $\mathbf{x}_{i,t}$ and $\mathbf{w}_{i,t}$ are allowed to load on the same set of factors as $e_{i,t}$ means that they can be endogenous. This type of endogeneity through unobserved heterogeneity is standard in panel data. Here we are considering interactive fixed effects, but the idea is the same in the fixed-effects case; the effects sitting in errors might also be in the regressors, which means that they have to be removed prior to estimation. In the fixed-effects case, one augments (1) by dummy variables, which is tantamount to transforming the variables into deviations from means. Here a more elaborate augmentation approach is needed, which is to be expected because interactive effects are more

3. We use the term “unobserved heterogeneity” although $\mathbf{f}_t$ might be known. Later, we elaborate on this.

4. Bai and Perron (1998) clarify that in the presence of nonstationary regressors, the break-date consistency and rate of convergence remain but not the limiting distributions. Ditzen, Karavias, and Westerlund (2025) conjecture that the factors can be nonstationary without any differences to the methodology.

5. The factor-in-regressors condition rules out lags of the dependent variable as regressors in the panel case. However, this is not really a restriction on the data-generating process, because general forms of serial correlation are still permitted through both factors and idiosyncratic errors. Therefore, one can fit a static model without the lagged dependent variable, which is left as serial correlation in the errors. The main limitations of doing this are 1) inefficiency because we do not account for the serial correlation in the estimation and 2) loss of interpretation in some models where the coefficient of the lagged dependent variable is the parameter of interest. Limitation 1 may not be important in moderate or large panels. Lagged dependent variables are not ruled out in the time-series case (see Bai and Perron [1998]).

general than fixed effects. Had the factors been known, which would be the case if the unobserved heterogeneity were made up of known deterministic terms, for example, we would have estimated

$$y_{i,t} = \mathbf{x}'_{i,t}\boldsymbol{\beta} + \mathbf{w}'_{i,t}\boldsymbol{\delta}_j + \mathbf{f}'_t\boldsymbol{\gamma}_i + \varepsilon_{i,t} \quad (5)$$

by ordinary least squares (OLS). This is possible because with $\mathbf{f}_t$ as a regressor, the regression error is no longer given by $e_{i,t}$ but by $\varepsilon_{i,t}$, which is independent of the regressors. On the other hand, if $\mathbf{f}_t$ is not known, then we need a good proxy to use in its stead.⁶ Ditzén, Karavias, and Westerlund (2025) use $\bar{\mathbf{x}}_t = N^{-1} \sum_{i=1}^N \mathbf{x}_{i,t}$ and $\bar{\mathbf{w}}_t = N^{-1} \sum_{i=1}^N \mathbf{w}_{i,t}$, and so do we.⁷ The appropriately augmented version of (1) is therefore given by

$$y_{i,t} = \mathbf{x}'_{i,t}\boldsymbol{\beta} + \mathbf{w}'_{i,t}\boldsymbol{\delta}_j + \bar{\mathbf{x}}'_t\mathbf{a}_i + \bar{\mathbf{w}}'_t\mathbf{b}_{i,j} + \text{error} \quad (6)$$

Because asymptotically observing the cross-sectional averages is just as good as observing the true factors, the regressors in (6) are asymptotically exogenous. This means that the estimation can be carried out using OLS. This is the same idea as in the common correlated effects (CCE) estimator of (Pesaran 2006), except that here we do not include the cross-sectional average of $y_{i,t}$ as a regressor in (6) (see Karavias, Narayan, and Westerlund [2023] for a discussion). If $\mathbf{f}_t$ is neither completely known nor completely unknown, as is usually the case in practice, then the cross-sectional averages will take care of the unknown factors, and the known factors can be added to (6) as additional regressors.⁸

If $N = 1$ such that (1) is a time-series model, then by definition, there is no cross-sectional variation that we can exploit to estimate unknown factors. Hence, in this case, $\mathbf{f}_t$ must be known, so the model to be fit is given by (5), which is then the same as in Bai and Perron (1998).

6. Take the so-called wage curve model, which relates worker's wages to the rate of unemployment and is the most common motivating example in the literature (see, for example, Bai [2009]). Here the factor loadings may represent workers' unobservable skills, such as innate ability, perseverance, and motivation, and the factors would represent the price of these skills, which is not necessarily constant over time.

7. The intuition for why argumentation by $\bar{\mathbf{x}}_t$ is needed is simple. First, we note that by (3), $\bar{\mathbf{x}}_t = \bar{\boldsymbol{\Gamma}}'_x \mathbf{f}_t + \bar{\mathbf{u}}_{x,t}$, where $\bar{\boldsymbol{\Gamma}}_x$ and $\bar{\mathbf{u}}_{x,t}$ are the cross-sectional averages of $\boldsymbol{\Gamma}_{x,i}$ and $\mathbf{u}_{x,i,t}$, respectively. Hence, provided that $\mathbf{u}_{x,i,t}$ is mean zero and independent across i , by a central limit law, $\bar{\mathbf{x}}_t \rightarrow_p \bar{\boldsymbol{\Gamma}}'_x \mathbf{f}_t$ as $N \rightarrow \infty$, where " $\rightarrow_p$ " signifies convergence in probability. We say that $\bar{\mathbf{x}}_t$ is "rotationally consistent" for $\mathbf{f}_t$, which is enough if the purpose is to control for $\mathbf{f}_t$. The intuition for why $\bar{\mathbf{w}}_t$ is needed is analogous.

8. In principle, the cross-sectional averages can capture all factors, whether or not they are observed (known). However, an important condition for this to work is that the number of cross-sectional averages is not fewer than the number of factors they replace. Because of this, it is a good idea to treat known factors as additional regressors in (6) because this makes it possible to estimate more unknown factors.

3 The toolbox

3.1 Hypothesis testing

This section presents the first set of tools, which is necessary for establishing that one or more structural breaks have happened and for determining their number, s . In particular, we consider tests of three hypotheses, labeled 1–3, and a sequential test to determine s . We begin by stating the hypotheses of interest.

1. H_0 : no breaks versus H_1 : s breaks, where the number of breaks under H_1 , s , is specified by the researcher.
2. H_0 : no breaks versus H_1 : $1 \leq s \leq s_{\max}$ breaks, where the maximum number of breaks under H_1 , $s_{\max}$, is specified by the researcher.
3. H_0 : s breaks versus H_1 : $s + 1$ breaks, where s is specified by the researcher.

To test hypotheses 1–3, we use several test statistics. The time-series versions of these tests have appeared in Bai and Perron (1998), while the panel versions have appeared in Ditzen, Karavias, and Westerlund (2025).

Hypothesis 1

If the dates of the breaks are known, we will consider a Chow test for hypothesis 1. Therefore, let us denote by $F(\mathcal{T}_s)$ the F statistic for testing the null of no breaks versus the alternative of s known breaks at dates $\mathcal{T}_s$, which is based on (5) in the time-series case and on (6) in the panel case. Appropriate critical values can be taken from the F distribution with s numerator degrees of freedom and $N\{T - p - (s + 1)q\} - p - (s + 1)q$ denominator degrees of freedom.

If $\mathcal{T}_s$ is unknown, which is most likely the case in practice, then the following supremum statistic can be used:

$$\sup F(s) = \sup_{\mathcal{T}_s \in \mathcal{T}_{s,\epsilon}} F(\mathcal{T}_s)$$

Here

$$\mathcal{T}_{s,\epsilon} = \{(T_1, \dots, T_s) : T_{j+1} - T_j \geq \epsilon T, T_1 \geq \epsilon T, T_s \leq (1 - \epsilon)T\}$$

is the set of permissible break dates, with ϵ being a user-defined trimming parameter. By setting $\epsilon \in (0, 1)$, we ensure that the breaks considered in the test are distinct and bounded away from the sample endpoints as assumed.

Hypothesis 2

Hypothesis 2 can be tested using the double maximum statistic

$$\text{WDmax}F(s_{\max}) = \max_{1 \leq s \leq s_{\max}} \frac{c_{\alpha,1}}{c_{\alpha,s}} \sup F(s)$$

where $c_{\alpha,s}$ is the critical value of $\sup F(s)$ at significance level α and s breaks. The weighting by $c_{\alpha,1}/c_{\alpha,s}$ here ensures that the marginal p -values of the weighted supremum statistics are all equal. This counterweights the decrease in the marginal p -value of $\sup F(s)$ that comes from increasing s and the resulting loss of power when s is large. The test statistic with weights $c_{\alpha,1}/c_{\alpha,s} = 1$ for all s is called $\text{UDmax}F(s_{\max})$.

Hypothesis 3

In section 4, we describe a procedure for how to estimate the break dates. Let $\hat{\mathcal{T}}_s = \{\hat{T}_1, \dots, \hat{T}_s\}$ be the set of estimated breaks obtained based on that procedure. For the test of hypothesis 3, we use the statistic

$$F(s+1|s) = \sup_{1 \leq j \leq s+1} \sup_{\tau \in \hat{\mathcal{T}}_{j,\epsilon}} F(\tau | \hat{\mathcal{T}}_s)$$

where $\hat{\mathcal{T}}_s$ contains estimates of the s break stipulated under the null hypothesis, τ is the additional $(s+1)$ th break under the alternative, and

$$\hat{\mathcal{T}}_{j,\epsilon} = \left\{ \tau : \hat{T}_{j-1} + (\hat{T}_j - \hat{T}_{j-1})\epsilon \leq \tau \leq \hat{T}_j - (\hat{T}_j - \hat{T}_{j-1})\epsilon, \hat{T}_0 = 0, \hat{T}_{s+1} = 1 \right\}$$

is the set of permissible breaks in between the estimated $(j-1)$ th and j th breaks. Hence, $F(s+1|s)$ is testing the null of s breaks versus the alternative that there is an additional break somewhere within the regimes stipulated under the null. Finally, $F(\tau | \hat{\mathcal{T}}_s)$ is the F statistic based on taking the estimated break dates in $\hat{\mathcal{T}}_s$ as given and testing for one additional break at τ .

The $F(s+1|s)$ test can be applied sequentially to estimate the number of breaks. In this case, we start by testing the null of no breaks against the alternative of a single break using $F(1|0)$. If the null is accepted, we set $\hat{s} = 0$ and terminate the procedure. However, if the null is rejected, we estimate the break point, denoted $\hat{T}_1$, and split the sample in two at $\hat{T}_1$. We then test for the presence of a break in each of the two subsamples using $F(2|1)$. If no breaks are found, we set $\hat{s} = 1$ and stop, whereas if breaks are detected, we estimate their location and split the sample again. This process continues until the test fails to reject.

The asymptotic distributions of the above tests in the pure time-series and panel cases can be found in Bai and Perron (1998) and Ditzen, Karavias, and Westerlund (2025), respectively. Because the distributions are the same, the critical values are the same. `xtbreak` therefore uses the critical values of Bai and Perron (1998, 2003), which are applicable for $\epsilon \in \{0.05, 0.1, 0.15, 0.2, 0.25\}$. In theory, the validity of the critical values requires $T \rightarrow \infty$ in the time-series case and $N, T \rightarrow \infty$ with $T/N \rightarrow 0$ in the panel case, which in practice means that T should be “large” in both cases and that N should be even larger in the panel case. For some Monte Carlo evidence on the accuracy of these predictions, we refer to Bai and Perron (2003) and Ditzen, Karavias, and Westerlund (2025).

4 Break-date estimation

The previous section was focused on testing for the existence of breaks and on determining their number. Now we turn to their location because researchers use break dates to identify the underlying cause of breaks.

The standard approach in the literature is to estimate breaks by minimizing the sum of squared residuals. Bai and Perron (1998) and Ditzen, Karavias, and Westerlund (2025) do the same. The break-date estimator included in `xtbreak` is therefore given by

$$\hat{\mathcal{T}}_s = \arg \min_{\mathcal{T}_s \in \mathcal{T}_{s,\varepsilon}} \text{SSR}(\mathcal{T}_s) \quad (7)$$

where $\text{SSR}(\mathcal{T}_s)$ is the sum of squared residuals based on s breaks. In the time-series case, the residuals are taken from (5), whereas in the panel case, they are taken from (6). If s is “small”, the minimization can be done by grid search. However, if s is “large”, then grid search, which requires $O(T^s)$ OLS operations, becomes computationally costly and possibly even infeasible. In such cases, the efficient dynamic programming algorithms of Bai and Perron (1998, 2003) and Ditzen, Karavias, and Westerlund (2025), which limit the number of operations to $O(T^2)$ for any s , can be used.

Once $\hat{\mathcal{T}}_s$ has been obtained, confidence intervals for each estimated break date can be constructed using the formulas given in Bai and Perron (1998) and Ditzen, Karavias, and Westerlund (2025).

5 The `xtbreak` command

5.1 Syntax

5.1.1 Automatic estimation of number of breaks and break dates

```
xtbreak depvar [indepvars] [if] [, options1 options2 options3 options5
options6]
```

This syntax tests for breaks via hypothesis 2 and estimates the number of breaks and break dates with no prior knowledge on number and location of breaks. Estimation of the number of breaks is based on the sequential test of hypothesis 3.

5.1.2 Testing for known structural breaks

```
xtbreak test depvar [indepvars] [if],
breakpoints(numlist, index|datelist, fmt(format)) [options1 options5]
```

This syntax implements hypothesis 1: testing for breaks if the break dates are known.

5.1.3 Testing for unknown structural breaks

```
xtbreak test depvar [indepvars] [if] [, hypothesis(1|2|3) breaks(real)
options1 options2 options3 options4 options5]
```

This syntax implements hypotheses 1–3: testing for breaks if the break dates are unknown.

The default is hypothesis 3 and option `sequential`.

5.1.4 Estimation of breakdates

```
xtbreak estimate depvar [indepvars] [if] [, showindex breaks(real) options1
options2 options5]
```

This syntax estimates break dates for a given number of breaks.

5.1.5 Maintenance

```
xtbreak[, update version]
```

The `update` option updates `xtbreak` from GitHub by using `net install`. The `version` option displays the version.

5.1.6 Specific options

options1 are general options and apply to `xtbreak` in general:

```
breakconstant noconstant nobreakvariables(varlist1) vce(type)
inverter(type) python
```

options2 are specific for unknown break dates:

```
trimming(real) error(real)
```

options3 are specific for unknown break dates and hypothesis 2:

```
wdmax level(#)
```

options4 are specific for unknown break dates and hypothesis 3:

```
sequential
```

options5 are specific to panel data:

```
breakfixedeffects nofixedeffects csd csa(varlist2) csanobreak(varlist3)  
kfactors(varlist4) nbkfactors(varlist5) noreweigh
```

options6 apply to the automatic estimation of the number and location of breaks:

```
skipsh2 cvalue(level) strict maxbreaks(#)
```

Data must be **tsset** or **xtset** (see [TS] **tsset** or [XT] **xtset**) before using **xtbreak**. Time-series data must not include gaps. Panel data can be unbalanced. In this case, observations with missing data will not be included in the regressions. *devar*, *indepvars*, and *varlist* may contain time-series operators; see [U] 11.4.4 **Time-series varlists**. **xtbreak** requires the **moremata** package (Jann 2005).

5.2 Description of options

xtbreak automatically detects if $N = 1$ or $N > 1$. Therefore, the user does not have to specify whether the data have a time series or a panel structure. **xtbreak** requires the **moremata** package (Jann 2005).

breakpoints(*numlist*, *index* | *datelist*, *fmt*(*format*)) specifies the known break points, which can be set by either the number of corresponding observations or the values of the time identifier. If *numlist* is used, option *index* is required. For example, **breakpoints**(10, *index*) specifies that one break occurs at the 10th observation as ordered by time. *datelist* takes a list of dates. For example, **breakpoints**(2010Q1, *fmt*(*tq*)) specifies a break in the first quarter of 2010. The option *fmt*() specifies the format and is required if *datelist* is used. The format set in **breakpoints**() and the time identifier need to be the same.

hypothesis(1|2|3) specifies which hypothesis to test. Specify **hypothesis**(1) for hypothesis 1, **hypothesis**(2) for hypothesis 2, and **hypothesis**(3) for hypothesis 3. The default is **hypothesis**(3) in combination with the option **sequential**.

showindex shows confidence intervals as the index.

breaks(*real*) specifies the number of unknown breaks under the alternative. For hypothesis 2, **breaks**() can include two values. For example, **breaks**(4 6) tests the null of no breaks against the alternative of 4–6 breaks. If only one value is specified, then the lower bound of the number of breaks under the alternative is set to 1. If hypothesis 3 is tested, then **breaks**() defines the number of breaks under the alternative. If hypothesis 3 is tested and **breaks**() is not defined, then option **sequential** is invoked.

breakconstant adds a break in the constant. The default is no breaks in the constant.

noconstant suppresses the constant.

nobreakvariables(*varlist*₁) defines variables with no structural breaks. *varlist*₁ can contain time-series operators.

vce(*type*) specifies the covariance matrix estimator. The options are **ssr** (homoskedastic errors, the default); **hc** (heteroskedasticity-robust); **hac** (heteroskedasticity- and autocorrelation-robust); **np** (the nonparametric estimator of Pesaran [2006]); and **wpn** (the fixed-*T* standard errors of Westerlund, Petrova, and Norkute [2019]).

inverter(*type*) sets the inverter. *type* can be **speed** (**invsym**()); **precision**; **qr** (equivalent to **precision**; **qrintv**()); **chol** (**cholinv**()); **p** (**pinv**()); or **lu** (**luinv**()). The choice of inverter has implications on speed and precision. For an overview, see [M-4] **Solvers**.

python uses Python to calculate segment-specific SSRs to improve speed. This option requires Stata 16 or later, Python, and the following packages: **scipy**, **numpy**, **pandas**, and **xarray**. Numerical differences in the calculations may occur because of different matrix inverters and precision in Stata and Python. This option can be used only with balanced panels.

trimming(*real*) specifies the trimming parameter as a percent. The trimming affects the minimal time periods between two breaks. The default is 15% (**trimming**(0.15)). Critical values are available for 5%, 10%, 15%, 20%, and 25%.

error(*real*) defines the error margin for the partial break model.

wdmax weighs the double maximum test statistic used for testing hypothesis 2. The default is to not use any weights.

level(*#*) sets the significance level for critical values for the double maximum test. If a value is chosen for which no critical values exist, **xtbreak test** will choose the closest level.

sequential specifies the sequential *F* test to obtain the number of breaks when using hypothesis 3.

breakfixedeffects adds breaks in individual fixed effects. The default is no breaks in fixed effects.

nofixedeffects suppresses individual fixed effects.

csd implements **csa**(*w*) **csanobreak**(*x*) automatically. For example, the variables in *w* would enter with breaks, while those in *x*, specified with the **nobreakvariables**() option, enter without breaks.

csa(*varlist*₂) specifies the variables with breaks, which are added as cross-sectional averages. **xtbreak** automatically calculates the cross-sectional averages.

csanobreak(*varlist*₃) is the same as **csa**() but for variables without breaks.

`kfactors(varlist4)` specifies that variables in `varlist4` are known factors, or variables in the data that are constant across the cross-sectional dimension. Examples are seasonal dummies or other observed common factors such as asset returns and oil prices. The factors in this list are affected by structural breaks in that their loadings change.

`nbkfactors(varlist5)` is same as above, but the factors in `varlist5` are not affected by structural breaks.

`noreweigh` can be applied only to unbalanced panels. By default, `xtbreak` reweighs the time-unit-specific errors used in the SSR by weights equal to total number of units in the sample divided by the number of unit observations in that period. Thus, it increases the SSR in segments with missing data. `noreweigh` avoids reweighting missing data.

`skiph2` skips hypothesis 2 (H_0 : no break versus H_1 : $0 < s < s_{\max}$ breaks) when running `xtbreak` without the `estimate` or `test` option.

`cvalue(level)` specifies the significance level to be used to estimate the number of breaks using the sequential test. For example, `cvalue(0.99)` uses the 1% significance level critical values to determine the number of breaks using the sequential test. See `level()` for further details.

`strict` enforces strict behavior of the sequential test to determine the number of breaks. The sequential test will stop once $F(s + 1|s)$ is not rejected given a rejection of $F(s|s - 1)$. This option improves speed in large time series but should be used with caution.

`maxbreaks(#)` limits the maximum number of breaks when using the sequential test.

5.3 Stored results

`xtbreak estimate` stores the following in `e()`:

Scalars

<code>e(num_breaks)</code>	number of breaks; if automatic detection is used, then this is the estimated number of breaks
----------------------------	---

Matrices

<code>e(breaks)</code>	matrix with break dates
<code>e(CI)</code>	confidence intervals with dimension $4 \times \text{e(num_breaks)}$; the first two rows are the lower and upper 95% intervals using time indices, and the second two rows are in the value of the time identifier
<code>e(SSRvec)</code>	vector with SSRs for selection of break dates; only available when <code>breaks(1)</code> is used
<code>e(SSRmat)</code>	matrix with segment-specific SSR; row indicates start, and column indicates end of segment

xtbreak test stores the following in **r()** for known break points:

Scalars	
r(Wtau)	value of test statistic
r(p)	<i>p</i> -value from <i>F</i> distribution

For unknown break points, the following is stored:

Scalars	
r(supWtau)	value of the $\sup F(s)$ statistic (hypothesis 1)
r(Dmax)	value of unweighted double maximum test statistic (hypothesis 2)
r(WDmax)	value of weighted double maximum test statistic (hypothesis 2)
r(f)	value of the $F(s+1 s)$ statistic (hypothesis 3)
r(c90)	critical value at the 90% level
r(c95)	critical value at the 95% level
r(c99)	critical value at the 99% level

5.4 Postestimation

The following postestimation commands can be used after **xtbreak estimate**:

estat indicator [*newvar*]

estat indicator creates an indicator variable $(1, \dots, 1, 2, \dots, 2, \dots, s+1, \dots, s+1)'$ that specifies all break regimes.

estat split [*varlist*]

estat split will split a *varlist* according to the estimated break points and draw a scatterplot of the variable with the break on the *x* axis and the dependent variable on the *y* axis, use

estat scatter *varname* [, *twoway_options*]

estat indicator creates a new variable of the form $(1, \dots, 1, 2, \dots, 2, \dots, s+1, \dots, s+1)'$, where the value changes for each break regime. **estat split** splits the variables defined in *varlist* according to the break dates and saves the names of the created variables in **r(varlist)**. **estat scatter** draws a scatterplot with the dependent variable on the *y* axis and a variable with breaks defined in *varname* on the *x* axis.

xtbreak estimate stores information about segment-specific SSRs and the SSRs for different break dates in **e()**. This information can be used to draw a time-series plot of SSRs across potential break dates with

estat ssr [, *tsline_options*]

estat ssr is available only after **xtbreak estimate** *y x*, **breaks(1)**. *tsline_options* are any options permitted when using **tsline**; see [TS] **tsline** or [G-2] **graph twoway tsline**.

5.5 On the choice of factors in the panel case

xtbreak is versatile in dealing with common factors, whether they are known (observed) or unknown (unobserved), through the options **csd**, **csa()**, **csanobreak()**, **kfactors()**, and **nbkfactors()**. Unknown factors are estimated by the cross-sectional averages specified in **csa()** and **csanobreak()** or, alternatively, in **csd**. Known factors can be dummy variables or other observed variables that do not vary across units and are defined by **kfactors()** and **nbkfactors()**. Whenever they are available, known factors should be included because they make it possible to estimate more unknown factors. These factors can be free of breaks, or they can be allowed to break.

Fixed effects are known factors but are not treated through the **kfactors()** and **nbkfactors()** commands so that **xtbreak**'s command structure can deal with fixed effects similarly to other Stata commands. In particular, one has to specify whether **xtbreak** should fit a model with or without overall constant or individual fixed effects separately from the factor structure specification. In total, **xtbreak** supports five different models, which are presented in table 1. The default is a model with fixed effects that are not breaking.

Table 1. Constant or fixed-effects model specifications

Specification	Model	Options
Individual fixed effects	$\alpha_i + e_{i,t}$	noconstant*
Fixed effects with breaks	$\alpha_{j,i} + e_{i,t}$	breakfixedeffects noconstant
Overall constant	$\alpha + e_{i,t}$	nofixedeffects
Constant with breaks	$\alpha_j + e_{i,t}$	breakconstant nofixedeffects
Nothing	$e_{i,t}$	noconstant nofixedeffects

NOTES: * is the default. Option **noconstant** is not necessary. $j = 1, \dots, s+1$. The remaining observed and unobserved factors are in $e_{i,t}$ and require use of **csd**, **csa()**, **csanobreak()**, **kfactors()**, and **nbkfactors()** as appropriate.

The choice of deterministic model has implications for the analysis of structural breaks. In the overall constant model, the constant is treated as a regular regressor, and we can test and detect breaks in it. In fact, the constant may be the only regressor. In the presence of fixed effects, **xtbreak** cannot be used without breaking regressors ($w_{i,t}$), because it cannot detect or estimate breaks that affect only fixed effects and not the regressors. Once breaking regressors are included, one has the choice to allow for breaking or nonbreaking fixed effects.

5.6 Unbalanced panels

xtbreak can be used with unbalanced panel data. Pure time-series data ($N = 1$) with gaps are not allowed. For unbalanced panels, there is an appropriate adjustment in the degrees of freedom in the test statistics. It is assumed that missing data are missing completely at random and that all time periods have at least one observation.

In terms of break-date estimation, `xtbreak` identifies breaks by minimizing the SSR; see (7). If a value is missing, Stata removes the whole unit-time observation, so the number of summands in the SSR drop, which can lead to estimating break dates away from the true break. To avoid this behavior, `xtbreak` reweighs the individual time-specific residuals $\epsilon_{i,t}$, and the SSR is calculated as

$$\text{SSR}_{\tau_S} = \sum_{t \in \tau_S} \sum_{i \in N_t} \left(\frac{N}{N_t} \epsilon_{i,t} \right)^2$$

where N is the number of total units and N_t is the number of units for which data are nonmissing in time period t . N/N_t scales the residuals up when data are missing. The option `noreweigh` avoids the reweighing by setting $N_t = N$ independently from the number of nonmissing observations.

6 COVID-19 deaths and cases

6.1 Time-series evidence for the whole US

Main results

In this section, we explain the use and options of `xtbreak`. We want to test whether we can identify structural breaks in the relationship between the number of COVID-19 deaths and cases in the US in 2020 and 2021. This is an interesting topic because COVID-19's case fatality rate, which is the number of deaths from COVID-19 over the number of COVID-19 cases, was a key variable of medical and policy interest and consequently was the focus of many studies; see, for example, Mathieu et al. (2020).⁹ The fatality rate can be time varying because of, for example, countrywide changes in the capability of detecting the virus, lockdowns, population-wide vaccination programs, improvements in treatment and hospital capacity, and the emergence of new strains. All of these events are massive in scale and can cause structural breaks. We use aggregate US weekly time-series data on the number of deaths and new cases from the Centers for Disease Control and Prevention. The dataset is also available on our GitHub page.

We want to fit the model

$$\Delta \text{deaths}_t = \beta_0 + \delta_1 \Delta \text{cases}_{t-1} + \delta_2 \Delta \text{cases}_{t-2} + \delta_3 \Delta \text{cases}_{t-3} + e_t \quad (8)$$

where deaths_t and cases_t are the reported deaths due to COVID-19 in week t and the number of new cases for the entire US in the same week, respectively. We assume that a week lies, on average, between a positive test and a possible death.¹⁰ The data range from 27 January 2020 (beginning of week 4) to 29 August 2021 (end of week 34). Δ denotes first differencing, which is taken to ensure stationarity.¹¹ The model is a

9. Case fatality rate statistics can be found here: <https://ourworldindata.org/mortality-risk-covid>.

10. Note that because we use weekly data, there can be up to 13 days between a positive test and a death. Monday is taken as the first day of a given week.

11. The regression (8) can be seen as the static $\Delta \text{deaths}_t = \beta_0 + \delta_1 \Delta \text{cases}_t + e_t$ model, where we use lags of Δcases_t to account for reverse causality concerns.

simplification, although it has been used elsewhere in the literature; see, for example, Silverio et al. (2020) and Fritz (2022).

We want to test whether the coefficients δ_1 , δ_2 , and δ_3 are subject to structural breaks. There are several reasons to believe that the relationship between the number of new cases and deaths might have changed. At the beginning of the pandemic, the understanding of COVID-19 and the best way to treat the disease were not developed. The numbers were also underreported because of limits in testing capacity and reporting routines. With better testing capacity, reporting routines, and treatments, the relationship is expected to change, especially after the first two waves, because of the knowledge gained. Starting in mid-December 2020, vaccines were introduced. Hence, the relationship between the number of cases and deaths might be expected to have changed again. Therefore, it seems reasonable to expect at least two breaks.

Figure 1 plots the number of deaths and cases over time. Note the striking difference between the number of deaths and cases in the first wave and the number of deaths being many times larger than the number of new cases. This difference is markedly smaller in the second wave, but the number of deaths is still higher. The third wave was the worst in terms of numbers by far, but the number of deaths per new case was much lower than before. From about week 29 in 2020, the number of cases started to pick up again, and so did the number of deaths.

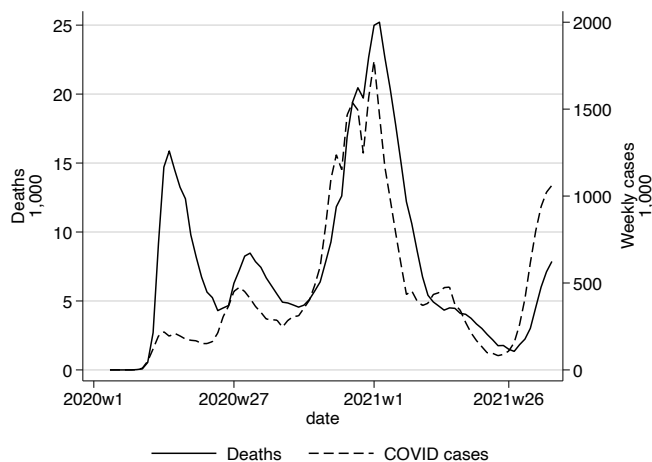


Figure 1. Plotting COVID-19 deaths and cases over time

We can use `xtbreak` without any prior knowledge of the number of breaks or their exact dates. We use the following command line:

```
. use us
. xtbreak d.deaths d.L(1/3).cases
```

Test for multiple breaks at unknown breakdates
(Bai & Perron. 1998. Econometrica)
H0: no break(s) vs. H1: 1 ≤ s ≤ 5 break(s)

	Test Statistic	Bai & Perron 1% Critical Value	Critical Values 5% Critical Value	10% Critical Value
UDmax	28.91	6.09	4.74	4.13

Sequential test for multiple breaks at unknown breakpoints
(Ditzen, Karavias & Westerlund. 2025)

	Test Statistic	Bai & Perron 1% Critical Value	Critical Values 5% Critical Value	10% Critical Value
F(1 0)	28.51	6.09	4.66	4.03
F(2 1)	5.47	6.59	5.24	4.64
F(3 2)	2.78	6.92	5.61	4.99
F(4 3)	2.70	7.33	5.87	5.23
F(5 4)	19.84	7.49	6.05	5.45

Detected number of breaks: (min)	1	2	2
(max)	5	5	5

Null hypothesis rejected more than once after non-rejection.
The detected number of breaks indicates the minimum and maximum
number of breaks for which the null hypothesis is rejected.

Estimation of break points

Number of obs	=	79
SSR	=	49.07
Trimming	=	0.15

#	Index	Date	[95% Conf. Interval]	
1	15	2020w22	2020w21	2020w23
2	45	2020w52	2020w51	2021w1

`xtbreak` starts with hypothesis 2; that is, it tests the null hypothesis of no breaks against the alternative of an unknown number of breaks between 1 and $s_{\max}$ breaks. This is the most powerful test, and it does not need knowledge of the number of breaks. The maximum number of breaks is set to $s_{\max} = 5$.¹² The value of the `UDmax` test statistic is 28.91, which is well above the 1% critical value and hence provides evidence for at least 1 and up to 5 breaks. `xtbreak` then estimates the number of breaks by

12. By default, the maximum number of breaks is set to the maximum number of breaks permissible by the trimming parameter. The maximum permissible number of breaks depends on the minimal length of the subsamples considered and hence on the trimming parameter ϵ . The maximum number of breaks is given by the formula $\lceil 1/\epsilon \rceil - 2$, where $\lceil \cdot \rceil$ is the smallest greater integer function. For $\epsilon = 0.15$, the permissible number of breaks is 5, while for $\epsilon = 0.10$, it is 8, and for $\epsilon = 0.05$, it is 18.

reporting the test value of hypothesis 3 (H_0 : s breaks versus $s + 1$ breaks) at each step in the sequence and the appropriate critical values for the three basic significance levels. We see that 0 break is rejected in favor of 1 break and that 1 break is rejected in favor of 2 breaks at the 5% significance level, but then, when testing 2 breaks against 3 or more breaks, the test is no longer able to reject.¹³ Therefore, we conclude that there are 2 breaks. `xtbreak` then proceeds to report the estimated break dates and the associated confidence intervals. The first break is estimated to week 22 of 2020 (fourth week of May), while the second is estimated at the end of year 2020. The confidence intervals on both breaks span only a couple of weeks, suggesting that the estimates are precise. The confidence interval width is a function of the magnitude of the break, and short confidence intervals hint at larger breaks.

`xtbreak` saves the values of the break points in `e(breaks)` and the confidence intervals in `e(CI)`. The matrices contain the index number $t \in \{1, \dots, T\}$ of both breaks and the confidence interval bounds. We use this information to draw figure 2, in which the estimated break dates and their 95% confidence intervals are plotted in the same graph as the deaths and the lagged number of cases.¹⁴

13. `xtbreak` does not stop and reports results for all tests up to the maximum number of breaks. This aims to provide a broader view of the sample properties because individual tests can suffer from type 1 errors. In this example, we see that the null hypothesis of four breaks against the alternative of five breaks is strongly rejected. This can be investigated in a robustness analysis, but we do not pursue it here. In such cases, one should keep an eye out on the magnitude of breaking coefficients because the method may be detecting small breaks that are not economically significant; see the empirical application in Ditzen, Karavias, and Westerlund (2025). An alternative explanation could be that there are multiple breaks with opposing magnitudes that “hide” each other until the number of breaks is set high enough to disentangle the opposing signs—and then the breaks become visible. This is why it is important to test hypothesis 2 before applying the sequential tests. However, this last scenario is unlikely in the current example because it is expected that the fatality rate only drops in time. The detected minimum and maximum numbers of breaks correspond to the first and last $F(s + 1|s)$ test rejections; at the 1% confidence level, the minimum detected number of breaks is 1 because $F(1|0)$ rejects, and the maximum is 5 because $F(5|4)$ rejects. It is possible to stop this process at the first nonrejection at the specified level using the option `strict`.

14. The code for the plot is available in the accompanying do-file.

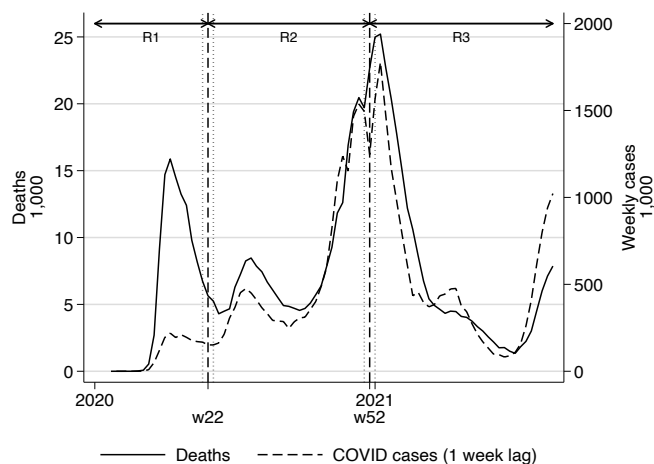


Figure 2. Plotting estimated breaks (dashed lines), 95% confidence intervals (dotted lines), deaths, and lagged cases

Next we use the `estat` command of `xtbreak` to generate new variables for each break regime and run an OLS regression:

```
. estat split
New variables created: LD_cases1 LD_cases2 LD_cases3 L2D_cases1 L2D_cases2
> L2D_cases3 L3D_cases1 L3D_cases2 L3D_cases3
. regress d.deaths `r(varlist)'
```

Source	SS	df	MS	Number of obs	=	79
Model	161.21305	9	17.9125611	F(9, 69)	=	25.19
Residual	49.0738642	69	.711215423	Prob > F	=	0.0000
				R-squared	=	0.7666
				Adj R-squared	=	0.7362
Total	210.286914	78	2.69598608	Root MSE	=	.84334

D.deaths	Coefficient	Std. err.	t	P> t	[95% conf. interval]
LD_cases1	.0625909	.009525	6.57	0.000	.043589 .0815928
LD_cases2	.0014608	.0015401	0.95	0.346	-.0016116 .0045332
LD_cases3	.0080629	.0012074	6.68	0.000	.0056542 .0104717
L2D_cases1	.0298979	.0121457	2.46	0.016	.005668 .0541279
L2D_cases2	.0037864	.0018352	2.06	0.043	.0001251 .0074476
L2D_cases3	-.0005085	.0012946	-0.39	0.696	-.0030911 .0020741
L3D_cases1	-.0171167	.0096604	-1.77	0.081	-.0363887 .0021553
L3D_cases2	.0065269	.001718	3.80	0.000	.0030995 .0099543
L3D_cases3	.0030002	.0011672	2.57	0.012	.0006717 .0053288
_cons	-.1963545	.1030999	-1.90	0.061	-.4020332 .0093243

```

. nlcom (Regime1: _b[LD_cases1] + _b[L2D_cases1] + _b[L3D_cases1])
> (Regime2: _b[LD_cases2] + _b[L2D_cases2] + _b[L3D_cases2])
> (Regime3: _b[LD_cases3] + _b[L2D_cases3] + _b[L3D_cases3])
      Regime1: _b[LD_cases1] + _b[L2D_cases1] + _b[L3D_cases1]
      Regime2: _b[LD_cases2] + _b[L2D_cases2] + _b[L3D_cases2]
      Regime3: _b[LD_cases3] + _b[L2D_cases3] + _b[L3D_cases3]

```

D.deaths	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Regime1	.0753721	.0089054	8.46	0.000	.0579178	.0928265
Regime2	.011774	.0020155	5.84	0.000	.0078238	.0157243
Regime3	.0105547	.0014136	7.47	0.000	.007784	.0133254

The long-run multipliers within each regime are given by $\delta_{LR,j} := \delta_{1,j} + \delta_{2,j} + \delta_{3,j}$ for $j = 1, 2, 3$. They are reported using the `nlcom` command, which also provides confidence intervals. The $\delta_{LR,j}$ capture the total effect in time that an additional 1,000 COVID-19 cases have on deaths in regime j . The estimate for the first regime $\hat{\delta}_{LR,1}$, covering the period from week 4 of 2020 to week 22 of 2020, suggests that for each additional 1,000 cases of COVID-19, 75 people died on average. According to figure 2, the end date of this regime coincides with the end of the first wave. This estimate is quite high, which can be explained by the relatively small number of tests conducted at that time. The estimated effect of the number of cases is much lower in the second regime, which stretches from week 23 of 2020 to week 51 of 2020. Only 12 out of an additional 1,000 infected died. The coefficient of the third regime, lasting from week 1 in 2021 to the end of the sample, is almost the same, where 11 out of an additional 1,000 infected died. This is probably in part due to the vaccination rollout, which began in week 51 of 2020.

Additional results and discussions

This section is divided in two: 1) additional test results and 2) additional estimation results. The aim is to demonstrate how the command can be used for specific tests and estimations, as opposed to the automatic way used above. We begin by considering additional test results, starting with a test of hypothesis 1. Specifically, we test the null hypothesis of no breaks against the alternative of a break in week 22 in 2020 and another break in week 52 in 2020 using the regular Chow F test. These are the results:

```

. xtbreak test d.deaths d.L(1/3).cases, hypothesis(1)
> breakpoints(2020w22 2020w52, fmt(tw))
Test for multiple breaks at known breakdates
(Bai & Perron. 1998. Econometrica)
H0: no breaks vs. H1: 2 break(s)

F          =          19.95
p-value (F) =          0.00

```

The test value is 19.95, which is way out in the critical region of the F distribution. Therefore, we can comfortably reject the null of no breaks. To test the same hypothesis but with unknown break dates, we specify `breaks(2)`. The results look as follows:

```
. xtbreak test d.deaths d.L(1/3).cases, hypothesis(1) breaks(2)
Test for multiple breaks at unknown breakdates
(Bai & Perron. 1998. Econometrica)
H0: no break(s) vs. H1: 2 break(s)
```

	Test Statistic	Bai & Perron 1% Critical Value	Critical Values 5% Critical Value	10% Critical Value
supF	19.95	4.82	4.00	3.58

```
Estimated break points: 2020w22 2020w52
Trimming: 0.15
```

The test value is identical to before, as is the conclusion to reject the null hypothesis, which is to be expected because the given break dates were set to the estimated breaks obtained earlier. The difference is that when the breaks are treated as unknown, the critical values come from a nonstandard distribution because the statistic is the supremum of F tests over all possible break dates because these are determined by the trimming parameter. These critical values are more “honest” than those used for the Chow test because they account for the fact that the breaks are unknown. As a part of the test results, `xtbreak` reports the estimated break dates used to construct the test, and we can see that they coincide with those obtained earlier.

We now test the null of no breaks against the alternative of up to five breaks. This is an example of a test of hypothesis 2, the results of which are presented here below. As expected, the null hypothesis is firmly rejected.

```
. xtbreak test d.deaths d.L(1/3).cases, hypothesis(2) breaks(5)
Test for multiple breaks at unknown breakdates
(Bai & Perron. 1998. Econometrica)
H0: no break(s) vs. H1: 1 <= s <= 5 break(s)
```

	Test Statistic	Bai & Perron 1% Critical Value	Critical Values 5% Critical Value	10% Critical Value
UDmax	28.91	6.09	4.74	4.13

```
Trimming: 0.15
```

Next we test the null of three breaks against the alternative of four breaks, which is an example of hypothesis 3. We use the options `hypothesis(3)` and `breaks(4)` to specify that there are four breaks under the alternative. The results presented below suggest that we are unable to reject the null, which is consistent with the estimated number of breaks.

```
. xtbreak test d.deaths d.L(1/3).cases, hypothesis(3) breaks(4)
Test for multiple breaks at unknown breakpoints
(Bai & Perron. 1998. Econometrica)
H0: 3 vs. H1: 4 break(s)
```

	Test Statistic	Bai & Perron 1% Critical Value	Critical Values 5% Critical Value	10% Critical Value
F(s+1 s)*	2.70	7.33	5.87	5.23

```
* s = 3
Trimming: 0.15
```

The option **sequential** repeats the hypothesis 3 test, sequentially starting from no breaks under the null up to the specified number in **breaks**. Setting **breaks(5)** returns the sequential test results reported earlier.

```
. xtbreak test d.deaths d.L(1/3).cases, hypothesis(3) breaks(5) sequential
```

As a final test exercise, we consider two changes to the model. We begin by investigating how the above results are affected if we allow for breaks in the constant. To do so, we use the sequential test but add the option **breakconstant**:

```
. xtbreak test d.deaths d.L(1/3).cases, breakconstant
```

To save space, we omit the output, but we briefly describe it; **xtbreak** finds five breaks. Note also that the options **hypothesis(3)** and **sequential** are the defaults, so we have omitted them from the command. We further investigate if the break is only in the constant and not in the number of cases. We keep the option **breakconstant** and move the variable **L.cases** to the option **nobreakvar(L.cases)**:

```
. xtbreak test d.deaths, breakconstant nobreakvar(d.L(1/3).cases)
```

Now **xtbreak** detects no breaks at the 5% significance level, meaning that the breaks are driven from changes in the slope coefficients.

We end this section with some comments on the estimation results. The break-date results reported earlier can be obtained using the option **xtbreak estimate**. The appropriate command line is

```
. xtbreak estimate d.deaths d.L(1/3).cases, breaks(2)
```

As an illustration of the fitted regression model, we can draw a scatterplot with different symbols for the observations within each regime. The plot can be created using **xtbreak estat**:

```
. estat scatter d.L.cases, ytitle("Change in deaths in 1,000s")
> xtitle("Change in cases in 1,000s") autolegend(pos(6) cols(3)) scheme(stsj)
> name(xtbreak_estat, replace)
```

The plot is displayed in figure 3. The different markers represent the observations within each regime, and the location of the observations is indicative of the strength of

the estimated linear relationships and of their slopes. The first regime is described by dots that appear in an almost vertical line. The second and third regimes are described by rhombi and squares, respectively, that mostly lie along the 45-degree line. In particular, we see that the slope in the first regime differs quite markedly compared with the other two, which confirms the findings from the OLS regression. *twoway_options* (see [G-3] *twoway_options*) can be passed through. Option `autolegend(legend_options)` automatically creates the legend labels for each segment. *legend_options* (see [G-3] *legend_options*) are further options passed to `legend()`, as done here to control the placement and number of columns of the legend.

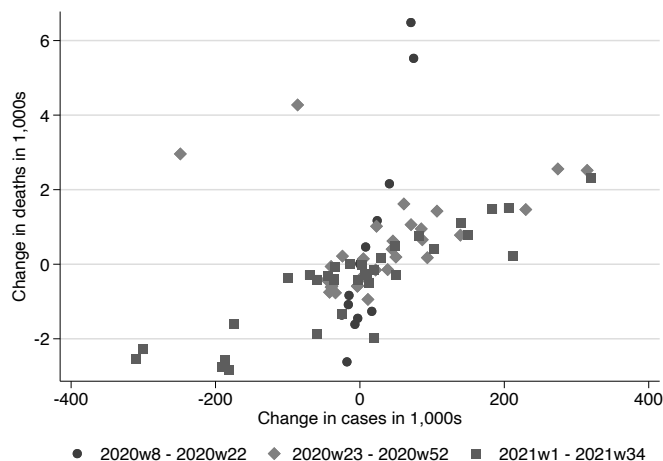


Figure 3. Scatterplotting lagged cases against deaths by regime

6.2 State-level panel evidence

Main results

In this section, we use the same US data as in the previous section; however, instead of aggregating the data up to the country level, we use data for all 50 US states, the District of Columbia, New York City, overseas territories, and three countries in free association with the United States.¹⁵

Like before, we report some results based on the sequential test to estimate the number of breaks and the break dates. We begin with the automatic `xtbreak` command that tests hypothesis 3 sequentially. The default panel-data case assumes no serial correlation or cross-sectional dependence. It has no constant but includes fixed effects. The distributed lag model used here captures serial correlation, so we use only

15. The overseas territories and three countries are American Samoa, Guam, the Commonwealth of the Northern Mariana Islands, Puerto Rico, the US Virgin Islands, the Federated States of Micronesia, Republic of the Marshall Islands, and Republic of Palau.

heteroskedasticity-robust standard errors. Additionally, we use a smaller trimming of 10%.¹⁶ `xtbreak` automatically detects whether a panel-data model is used, and thus the syntax remains the same. These are the results:

```
. use us_panel, clear
. xtbreak d.deaths d.L(1/3).cases, vce(hc) trimming(0.1)
```

Test for multiple breaks at unknown breakdates
(Ditzen, Karavias & Westerlund. 2025)

H0: no break(s) vs. H1: $1 \leq s \leq 9$ break(s)

	Test Statistic	Bai & Perron 1% Critical Value	Critical Values 5% Critical Value	10% Critical Value
UDmax	13.60	6.25	4.95	4.42

Sequential test for multiple breaks at unknown breakpoints
(Ditzen, Karavias & Westerlund. 2025)

		Bai & Perron Critical Values		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
F(1 0)	11.90	6.24	4.87	4.26
F(2 1)	9.31	6.78	5.51	4.85
F(3 2)	8.50	7.20	5.81	5.21
F(4 3)	10.76	7.45	5.99	5.49
F(5 4)	1.14	7.65	6.20	5.65
F(6 5)	4.82	7.79	6.34	5.78
F(7 6)	1.24	7.84	6.42	5.89
F(8 7)	2.20	7.90	6.54	5.98
F(9 8)	2.44	7.93	6.65	6.12

Detected number of breaks:	4	4	4
----------------------------	---	---	---

The detected number of breaks indicates the highest number of breaks for which the null hypothesis is rejected.

Estimation of break points

Number of obs	=	4740
Number of Groups	=	60
Obs per group	=	79
SSR	=	13.88
Trimming	=	0.10

#	Index	Date	[95% Conf. Interval]	
1	7	2020w14	2020w13	2020w15
2	14	2020w21	2020w20	2020w22
3	46	2021w1	2020w52	2021w2
4	53	2021w8	2021w7	2021w9

16. Thicker trimming offers better small-sample properties for the test statistics. The availability of rich datasets allows the use of thinner trimming, which permits more breaks and also allows breaks to be closer to each other. More breaks do not increase the computational burden per se because of the dynamic programming algorithms used. However, more breaks can mean more sequential tests of hypotheses $F(s+1|s)$, which will ultimately delay the automatic version of `xtbreak`. Limits on the number of `hypothesis(3)` tests can be introduced by using `xtbreak test` directly and by adding options such as `strict` and `maxbreaks()`.

At a 5% significance level, we find four breaks that are estimated in weeks 14 and 21 of 2020 and in weeks 1 and 8 of 2021. The second and third breaks are remarkably close to the two breaks found in the single-time-series analysis above.¹⁷ We will comment on the break dates below. Given that COVID-19 waves impact multiple states at the same time, there may be dependence across states. We address this issue by augmenting the model with the cross-sectional average of the lagged number of differenced cases.¹⁸ Additionally, we use the option `strict`, which uses the sequential test until it does not reject the null at the required significance level. `strict` provides the consistent number of breaks estimator of theorem 3.2 in Ditzen, Karavias, and Westerlund (2025). We also use the `skip2` option, which skips the test of no breaks against $1 < s < s_{\max}$ breaks to save space because it strongly rejects. We use the heteroskedasticity- and autocorrelation-robust variance estimator of Westerlund, Petrova, and Norkute (2019) because of its excellent small-sample properties. The results look as follows:

```
. xtbreak d.deaths d.L(1/3).cases, csa(d.l.cases) vce(wpn) trimming(0.1) skip2
```

```
Sequential test for multiple breaks at unknown breakpoints
```

```
(Ditzen, Karavias & Westerlund. 2025)
```

	Test Statistic	Bai & Perron 1% Critical Value	Critical Values 5% Critical Value	10% Critical Value
F(1 0)	6.32	6.24	4.87	4.26
F(2 1)	68.68	6.78	5.51	4.85
F(3 2)	73.44	7.20	5.81	5.21
F(4 3)	9.10	7.45	5.99	5.49
F(5 4)	4.50	7.65	6.20	5.65
F(6 5)	13.82	7.79	6.34	5.78
F(7 6)	10.33	7.84	6.42	5.89
F(8 7)	10.40	7.90	6.54	5.98
F(9 8)	8.04	7.93	6.65	6.12
Detected number of breaks: (min)		4	4	4
(max)		9	9	9

```
Null hypothesis rejected more than once after non-rejection.
```

```
The detected number of breaks indicates the minimum and maximum  
number of breaks for which the null hypothesis is rejected.
```

17. The model then can be fit by using the commands `estat split` to generate the breaking variables and `xtreg d.deaths `r(varlist)', fe` to run a fixed-effects regression.

18. The CCE methodology requires that the cross-sectional average of each regressor is used, which here translates to the cross-sectional averages of the three lags. This can be done by option `csd` or explicitly by `d.l(1/3).cases`. However, we point out that the model used is a distributed lag model with high persistence in the regressors. Thus, the three cross-sectional averages introduced by `d.l(1/3).cases` are highly correlated, containing almost the same information $\bar{x}_{t-1} \approx \bar{x}_{t-2} \approx \bar{x}_{t-3}$. In this case, it makes sense to use only one cross-sectional average, which we implement with the option `csa(d.l.cases)`.

Estimation of break points				
		Number of obs	=	4740
		Number of Groups	=	60
		Obs per group	=	79
		SSR	=	10.89
		Trimming	=	0.10
#	Index	Date	[95% Conf. Interval]	
1	7	2020w14	2020w13	2020w15
2	14	2020w21	2020w20	2020w22
3	45	2020w52	2020w51	2021w1
4	53	2021w8	2021w7	2021w9
Cross-section averages:				
with breaks: LD.cases				

The option `strict` has a default significance level of 5%. For this significance level, the $F(5|4)$ test does not reject, and the sequential procedure stops yielding an estimate of four breaks. Notably, the breaks are the same as before.

Additional results and discussions

The estimated breaks take place in week 14 of 2020, week 21 of 2020, week 52 of 2020, and week 8 of 2021. The confidence intervals for all four breaks are narrow. When we compare these results with those reported earlier for the time-series dataset for the whole US, we see that the second and third breaks coincide. Hence, for two of the break dates, the panel-data evidence reinforces the time-series evidence reported earlier. However, the panel-data results also suggest that two breaks are not enough and that there is a need to account for an early break in week 14 of 2020 and for a fourth break in week 8 of 2021. The fact that the panel toolbox detects two breaks that are not picked up by the time-series analysis could be due to the gain in accuracy obtained by using the larger panel dataset (see Ditzen, Karavias, and Westerlund [2025]).

The command `estat split` generates the breaking regressors that will be used as independent variables. The model is fit by CCE using `xtccce2` (Ditzen 2018, 2021) because the CCE estimator allows for interactive effects that capture cross-section dependence. As with the above, we report the long-run multipliers for each regime. The estimated multiplier in the first regime, which corresponds to the period of the first wave, is the highest because there was little medical knowledge about the virus at that time, limited preparedness, and limited capacity in detecting cases. The last can be seen in figure 4. The multipliers drop in the second regime as the first wave dissipates and more testing becomes available. The third regime includes the second and part of the third wave appearing in October 2020. The multiplier further drops here. The fourth regime includes the peak of the third wave. Here cases and deaths are close to each other because of extensive tracing programs used across states and also mass vaccinations. The multiplier remains almost the same in the fifth and final regime. This

is an example of a break that is statistically significant but may not be economically significant, so we do not consider more breaks in the analysis.¹⁹

```

. quietly xtdcce2 d.deaths `r(varlist)', pooled(`r(varlist)') cr(d.L.cases)
> pooledvce(wpn)
. nlcom (Regime1: _b[LD_cases1] + _b[L2D_cases1] + _b[L3D_cases1])
> (Regime2: _b[LD_cases2] + _b[L2D_cases2] + _b[L3D_cases2])
> (Regime3: _b[LD_cases3] + _b[L2D_cases3] + _b[L3D_cases3])
> (Regime4: _b[LD_cases4] + _b[L2D_cases4] + _b[L3D_cases4])
> (Regime5: _b[LD_cases5] + _b[L2D_cases5] + _b[L3D_cases5]), post
    Regime1: _b[LD_cases1] + _b[L2D_cases1] + _b[L3D_cases1]
    Regime2: _b[LD_cases2] + _b[L2D_cases2] + _b[L3D_cases2]
    Regime3: _b[LD_cases3] + _b[L2D_cases3] + _b[L3D_cases3]
    Regime4: _b[LD_cases4] + _b[L2D_cases4] + _b[L3D_cases4]
    Regime5: _b[LD_cases5] + _b[L2D_cases5] + _b[L3D_cases5]

```

D.deaths	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Regime1	.1331224	.0244026	5.46	0.000	.0852942	.1809506
Regime2	.0555998	.0163378	3.40	0.001	.0235783	.0876213
Regime3	.0150175	.0011888	12.63	0.000	.0126876	.0173475
Regime4	.0104796	.0015258	6.87	0.000	.007489	.0134701
Regime5	.0107557	.0010314	10.43	0.000	.0087343	.0127772

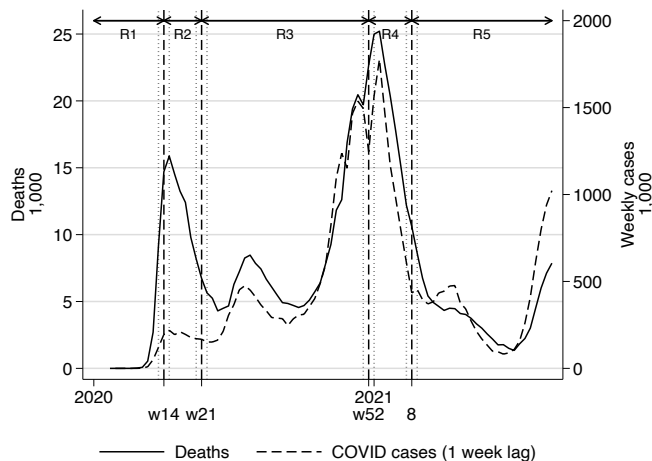


Figure 4. Plotting estimated breaks (dashed lines), 95% confidence intervals (dotted lines), deaths, and lagged cases. The arrows on top indicate the regimes.

19. In large datasets, an overspecified model with more breaks is preferable to an underspecified model because the estimators are inconsistent in the latter (see Ditzen, Karavias, and Westerlund [2025]).

7 Consumer confidence and leader approval rating

Another relationship that can potentially suffer from structural breaks is that between consumer confidence and the approval rating of a country's leader. In the US, there is now a large literature analyzing the determinants of presidential ratings (see, for example, Berlemann and Enkelmann [2014] for an extensive review). The determinant of interest here is consumer confidence, a statistical measure of consumer feelings on the state of the economy and their financial situation. Confidence in the economy should translate into high presidential ratings, although the relationship itself may be unstable at times with structural breaks occurring for various reasons, including political instability and other significant events, an example being the inauguration of President Barack Obama (Small and Eisinger 2020). Another dimension of interest when studying such relationships is their cross-country heterogeneity, which is the outcome of unique historical processes shaping institutions and culture. In the words of Nannestad and Paldam (1994), "approval relationships have shown a disappointing lack of stability both over time and across countries."

The present study is motivated by the above discussion and will attempt to discover breaks in a set of eight countries in which leader approval ratings were observed monthly from January 1990 to December 2021. The data on approval ratings were obtained from the EAP 3.0 database (Carlin et al. 2023), while the data on consumer confidence were taken from the OECD.²⁰ The two variables are denoted as **approval** and **CCI**. Because the time dimension is long, we set the trimming parameter to 5%. Additionally, we control for breaks because of elections, which can be seen as known break dates. Election dates for the years after 2000 come from the International Foundation for Electoral Systems (2023) and are entered manually for the years before 2000. To account for the political environment right before and after an election, the dummy variable **ElectionQ** equals 1 in the month before an election and for the two months after an election. We analyze first the whole panel and then each country separately.

In the estimations below, we allow unobserved heterogeneity to account for country-specific variation and use first differences of the variables to ensure stationarity. Because the time dimension is large, many possible SSRs must be estimated, which takes a considerable amount of time.²¹ To save time, we restrict the number of breaks under the null hypothesis for hypothesis 2 to $s_{\max} = 5$ using the options **maxbreaks(5)** and **strict** for the sequential test and invoke the option **python**.²² With the **strict** option, the sequential test will stop once the null hypothesis $F(s+1|s)$ is not rejected given a rejection of $F(s|s-1)$ at the default 5% significance level.

20. In the panel case, the validity of critical values theoretically requires that $N > T$. However, Monte Carlo evidence in Ditzen (2021) shows that this requirement is not critical. See also the discussion at the end of section 3.1.

21. The trimming of 5% and the 383 time periods implies a minimal segment length of $h = 20$ periods. The number of possible segments is $T(T+1)/2 - (h-1)T + (h-2)(h-1)/2 - h^2m(m+1)/2$ with m being the number of breaks (Bai and Perron 2003). For $m = 1$, this amounts to over 65,000 estimations.

22. We did not find more than five breaks for the entire panel or any individual country. Using the option **python** reduces the computation time significantly.

```
. use approval_panel, clear
. xtbreak d.approval d.CCI, trimming(0.05) nobreakvariables(ElectionQ) strict
> maxbreaks(5) python
```

Test for multiple breaks at unknown breakdates

(Ditzen, Karavias & Westerlund. 2025)

H0: no break(s) vs. H1: $1 \leq s \leq 5$ break(s)

	Test Statistic	Bai & Perron 1% Critical Value	Critical Values 5% Critical Value	10% Critical Value
UDmax	13.91	13.74	10.17	8.78

Sequential test for multiple breaks at unknown breakpoints

(Ditzen, Karavias & Westerlund. 2025)

	Test Statistic	Bai & Perron 1% Critical Value	Critical Values 5% Critical Value	10% Critical Value
F(1 0)	12.63	13.58	9.63	8.02
F(2 1)	26.99	15.03	11.14	9.56
F(3 2)	5.32	15.62	12.16	10.45

Detected number of breaks:	2	2	2
----------------------------	---	---	---

The detected number of breaks indicates the highest number of breaks for which the null hypothesis is rejected.

Estimation of break points

Number of obs	=	3064
Number of Groups	=	8
Obs per group	=	383
SSR	=	34584.14
Trimming	=	0.05

#	Index	Date	[95% Conf. Interval]	
1	344	2018m9	2017m3	2020m3
2	363	2020m4	2019m2	2021m6

The UDmax test is rejected at the 10% level, suggesting at least one break in the relationship. Such evidence is rather weak given the size of the sample. However, looking at the sequential statistic results, there is strong evidence in favor of two breaks at all significance levels. The two breaks are found in September 2018 and April 2020. In both cases, the confidence intervals of the breaks span more than one year, having a width of 36 and 28 months, respectively. The first break could be due to Donald Trump's election in the US; its confidence interval includes almost his entire first presidency. Trump's election is significant because leaders across the board try to side with or oppose his rhetoric and policies. The second break could be due to the COVID-19 pandemic, which was a difficult and multifaceted problem for leaders and the economy.

Table 2 presents some further robustness results examining how the number and location of breaks vary across model specifications. The parameters that vary are trimming, the constant or fixed-effects specification, the significance level, and whether "Elections" are included as the nonbreaking regressor. Overall, we observe that the panel results are robust to these choices.

Table 2. Leader approval rating and consumer confidence: Estimated breakpoints and confidence intervals

Break #	Individual fixed effects			Overall constant	
	(1)	(2)	(3)	(4)	(5)
1	2018m9 (2017m3, 2020m3)	2018m9 (2017m3, 2020m3)	2018m9 (2017m3, 2020m3)	2018m9 (2017m3, 2020m3)	2018m9 (2018m6, 2018m12)
2	2020m4 (2018m10, 2021m10)	2020m4 (2018m10, 2021m10)	2020m4 (2019m2, 2021m6)	2020m4 (2019m2, 2021m6)	2020m4 (2020m2, 2020m6)
UDmax	15.60***	9.73*	13.91***	13.85***	8.13***
Break in constant	No	No	No	No	Yes
Nonbreaking v.	-	-	Election	Election	Election
Trimming	5%	5%	5%	5%	5%
Critical v.	1%	5%	1%	1%	1%

NOTES: 95% confidence intervals in parentheses. UDmax is the test statistic of hypothesis 2 with no breaks against $1 < s < s_{\max} = 19$. Stars indicate significance level at ***1%, **5%, *10%. Level critical values indicate the critical values at which the break dates are selected. “Nonbreaking v.” indicates nonbreaking variable. “Critical v.” indicates critical variable.

We now explore cross-country heterogeneity by analyzing the disaggregated data for each of the eight countries in the sample. Individual country analysis allows for heterogeneous intercept and slope regression coefficients, although it can be less efficient. The results are depicted in figure 5, which includes the estimated break dates marked by vertical lines because they have been estimated in each time series. For comparison, the estimated breaks of the full panel are indicated by a dotted line. The number of estimated breaks ranges from none (Australia) to three (Germany). The April 2020 break estimated in the panel appears in five of eight countries: France, Germany, Spain, the UK, and the US. The break in 2018 appears only in Spain and the UK. Overall, the figure shows that the heterogeneity across countries is significant.

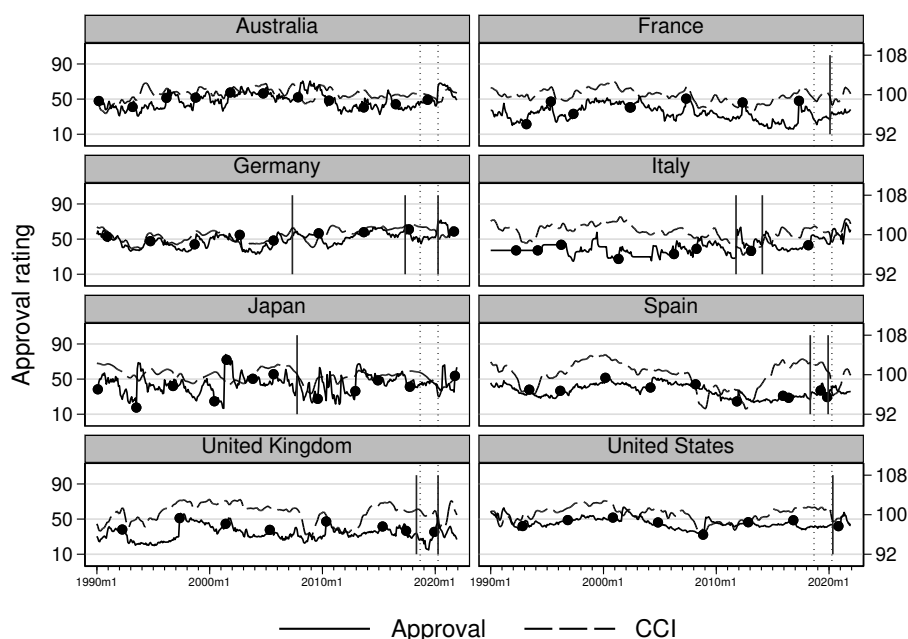


Figure 5. Leader confidence. Dashed lines indicate break-point estimates on country level; dotted lines indicate the same on panel level; and dots indicate elections.

As a final exercise, we return to the panel case and assume that there is a single break. We now plot the SSRs via `estat ssr` after using `xtbreak estimate`. The `estat ssr` command is available only after the estimation of a single break and displays the SSR for each possible break date.²³

23. The SSR function is different for different numbers of breaks. If there are two breaks, the SSR function becomes a surface.

```
. xtbreak estimate d.approval d.CCI, trimming(0.05) nobreakvariables(ElectionQ)
> breaks(1) python
```

Estimation of break points

```
Number of obs      =    3064
Number of Groups   =      8
Obs per group      =    383
SSR                =   34907.31
Trimming           =     0.05
```

#	Index	Date	[95% Conf. Interval]	
1	360	2020m1	2015m3	2024m11

```
. estat ssr, scheme(stsj) name(ssr, replace)
```

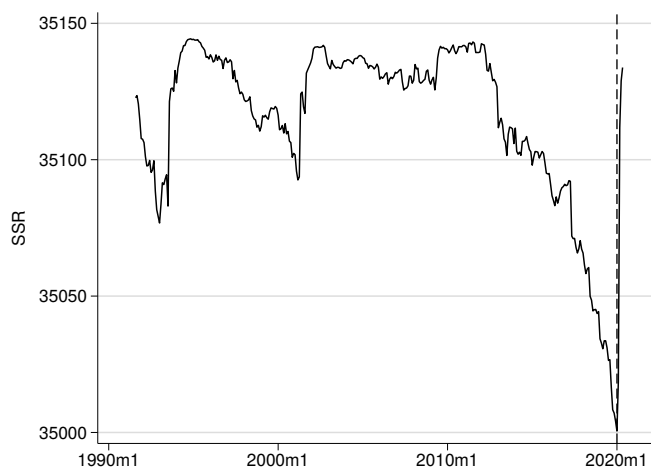


Figure 6. SSRs over time

The estimated break point is indicated by a dashed line. The break is estimated to be in January 2020, and the SSR of 34907.31 represents the sum of the SSR of an estimation ranging from January 1990 until January 2020 and the SSR of an estimation from February 2020 to December 2021. This break is almost identical to the second break in the panel-data model above and coincides with the second break appearing in five out of the eight countries of the sample. In an underspecified model in which the assumed number of breaks is less than the true number of breaks, the break-date estimators are still consistent for the true break dates (see Bai and Perron [1998]). Therefore, the estimator of the one break here is meaningful and converges to one of the true break points; which of the true breaks it will converge to depends on the magnitudes of the breaks and the duration of the regimes.

8 Conclusion

This article presents a new community-contributed command called `xtbreak`, which enables researchers to detect breaks and estimate their number and location. The command can be applied to time series and panel data and can hence be seen as a complete break detection toolbox that is applicable regardless of the structure of the data. The `xtbreak` command is applied to detect breaks in the relationship between COVID-19 cases and deaths and in the relationship between consumer confidence and the approval ratings of country leaders.

In the following, we discuss challenges, limitations, and further research ideas that are relevant to the current contribution. Computational speed can be a challenge, especially in datasets with many time-series observations. The dynamic programming algorithm embedded in `xtbreak` goes far in addressing this issue because it ensures that at most $O(T^2)$ calculations are made when the number of breaks is more than two; however, this number can still be large. `xtbreak` includes several options that can further improve computational speed, such as the use of Python or the use of faster but less precise matrix inversion methods. Speed gains can also be achieved through modeling choices: the pure structural-break model computes significantly faster than the partial break model. Consequently, it may be advantageous to assume that breaks affect all coefficients rather than just a subset. While assuming a coefficient break when none exists does not lead to inconsistent estimation, it does result in inefficiency; therefore, this option should be exercised with caution. Future research could explore additional speed improvements through multiprocess parallelization, which is something not explored thus far.

The panel-data methodology used in this article assumes common coefficients and common breaks across units. This is the prevalent setting in the applied literature; however, there are situations where heterogeneity is a significant feature of the data. Recently, steps have been taken to relax these assumptions; see, for example, Okui and Wang (2021), who propose a model in which different groups of units suffer a different number of breaks at different times. This method has not been coded in Stata so far, and it would be an important complement to the methodology in this article.

9 Acknowledgments

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10 Programs and supplemental material

To install the software files as they existed at the time of publication of this article, type

```
. net sj 25-3
. net install st0781      (to install program files, if available)
. net get st0781          (to install ancillary files, if available)
```

The latest version of the `xtbreak` package can be obtained by typing the following:

```
. net from https://janditzen.github.io/xtbreak/
. net install xtbreak
```

Updates and other details can be found at <https://janditzen.github.io/xtbreak/>.

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